

1. Given data

Parameter	Symbol	Value
Total vehicle mass	m	20 kg
Vehicle length	L	700 mm
Vehicle width	B	340 mm
Vehicle height	H	350 mm
Ground clearance	GC	50 mm
Hopper mass	m_h	8 kg
Roller mass	m_r	3 kg
Battery mass	m_β	1 kg
Number of drive wheels	n	2
Wheel diameter	D	100 mm
Wheel radius	r	50 mm = 0.05 m
Motor voltage	V	12 V

Assumption: Vehicle moves on asphalt/concrete at low speed (0.3–0.4 m/s).

2. Total vehicle weight

The gravitational force acting on the vehicle is:

$$W = m \times g$$

Where: $m = 20 \text{ kg}$, $g = 9.81 \text{ m/s}^2$

$$W = 20 \times 9.81$$

$$W = 196.2 \text{ N}$$

Total vehicle weight = 196.2 N

3. Rolling resistance

For rubber wheels on concrete:

$$\mu_r = 0.03$$

Rolling resistance force:

$$F_r = \mu_r \times W$$

$$F_r = 0.03 \times 196.2$$

$$F_r = 5.886 \text{ N}$$

Rolling resistance = 5.89 N

4. Starting resistance

The vehicle carries an 8 kg hopper and 3 kg roller, therefore additional force is required during startup.

Assume 12% of vehicle weight.

$$F_s = 0.12 \times 196.2$$

$$F_s = 23.54 \text{ N}$$

Starting resistance = 23.54 N

5. Total tractive force

$$F_t = F_r + F_s$$

$$F_t = 5.89 + 23.54$$

$$F_t = 29.43 \text{ N}$$

Required tractive force = 29.43 N

6. Wheel diameter selection

Ground clearance is 50 mm.

For a wheel:

$$GC = \frac{D}{2} - R_{roller}$$

Using:

- Roller radius = 36 mm
- Wheel diameter = 100 mm

$$Wheel\ radius = 50\ mm$$

This provides the required clearance while keeping the chassis compact.

Selected wheel

Parameter	Value
Diameter	100 mm
Width	35 mm
Material	Rubber
Hub bore	6 mm

7. Wheel torque calculation

$$\tau = rF\sin(\theta)$$

$$\tau = (3)(6)\sin(90^\circ) = 18\ N\ m$$

Wheel radius:

$$r = 0.05\ m$$

Torque required at wheels:

$$T = F_t \times r$$

$$T = 29.43 \times 0.05$$

$$T = 1.4715\ N \cdot m$$

Total wheel torque = 1.47 N·m

8. Torque per motor

Two motors share the load equally.

$$T_m = \frac{1.4715}{2}$$

$$T_m = 0.736 \text{ N} \cdot \text{m}$$

Convert to kg·cm:

$$1 \text{ N} \cdot \text{m} = 10.197 \text{ kg} \cdot \text{cm}$$

$$T_m = 0.736 \times 10.197$$

$$T_m = 7.50 \text{ kg} \cdot \text{cm}$$

Required torque per motor = 7.5 kg·cm

9. Apply safety factor

For mobile robots:

$$SF = 2$$

Design torque:

$$T_d = 7.5 \times 2$$

$$T_d = 15 \text{ kg} \cdot \text{cm}$$

Final design torque: 15 kg·cm minimum per motor

10. Motor speed calculation

Desired speed = 0.35 m/s

Wheel circumference:

$$C = \pi D$$

$$C = 3.1416 \times 0.1$$

$$C = 0.314 \text{ m}$$

RPM:

$$RPM = \frac{v \times 60}{C}$$

$$RPM = \frac{0.35 \times 60}{0.314}$$

$$RPM = 66.9$$

Nearest standard motor: 60 RPM

11. Motor verification

The Johnson motor specification:

Parameter	Value
Voltage	12 V
Rated speed	60 RPM
Rated torque	157.6 N·cm
Stall torque	850 N·cm

Convert rated torque:

$$157.6 \text{ N} \cdot \text{cm} = 1.576 \text{ N} \cdot \text{m}$$

Required:

$$1.47 \text{ N} \cdot \text{m}$$

Available:

$$1.576 \text{ N} \cdot \text{m}$$

Safety margin:

$$\frac{1.576}{1.47} = 1.07$$

The motor satisfies the continuous torque requirement.

Conclusion

The selected 12 V, 60 RPM Johnson geared DC motor with 157.6 N·cm rated torque is suitable for your 20 kg automated pothole measurement and repair system when used with 100 mm diameter wheels. The calculated required wheel torque is 1.47 N·m, and the motor provides 1.576 N·m, making it adequate for continuous operation at low speed.